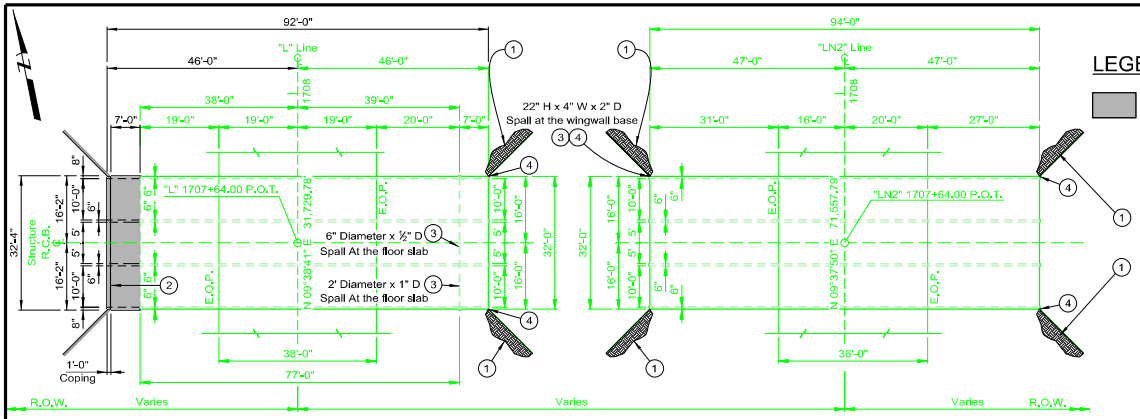
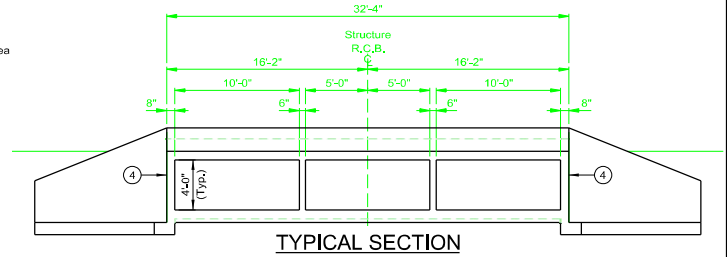
STATE	PROJECT NO.	COUNTY	SHEET NO.
NEVADA	SPF-095-1-(341)	CLARK	B6



PLAN

LEGEND

Remove & Replace Existing Structure Area



TYPICAL SECTION

DESCRIPTION OF WORK

(1) Repair Erosion
- A. Fill eroded areas at the wingwalls with Granular Backfill (Bid Item 207 0110).

(2) RCB Extension:
- A. Remove existing 6' extension, footing, and headwall (Bid Items 202 0125 and 206 0110). Saw cut (no direct payment) and remove 1' of existing RCB (Bid Item 202 0125 and 206 0110).
- B. Over-excavate (Bid Item 206 0110) 48" down and construct subgrade stabilization pad (Bid Items 203 0690, 203 0720, and 610 0470).
- C. Construct new triple 10' x 4' reinforced concrete box extension (Bid Items 502 0970, 505 0100 and 505 0110).
- D. The new box culvert extension and wingwalls shall be founded on 4" of sand bedding material and be constructed on a "Construction Platform"consisting of Riprap Bedding (Class 300) wrapped in Nonwoven Geotextile (Class 1) (Bid Items 203 0690, 203 0720 and 610 0470).

(3) Repair the spall and delamination at the wingwalls and barrels (Bid Item 502 0390).
- A. Remove damaged concrete to such depth that sound concrete is exposed over the entire area.
- B. Blast-clean any exposed reinforcing steel.
- C. Apply bonding adhesive and place concrete patch.
- D. Perform the above repairs in accordance with subsection 502.03.15 of the standard specifications.
- E. After the concrete has cured, apply finish to match the existing bridge color, no direct payment.

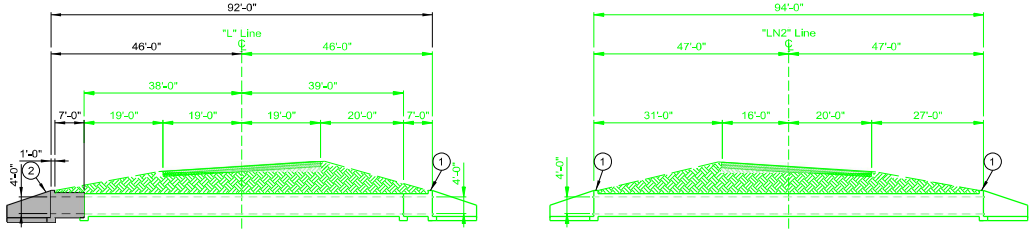
(4) Repair Cracks at the Headwall-Wingwall Juncture (Bid Item 502 0405).
- A. Clean cracked surface, remove loose materials, and clean with compressed air.
- B. Seal the cracks using epoxy injections with epoxy conforming to 728.03.01 of the Standard Specifications.
- C. Do not begin repairs until written procedure and repair material is approved by the Engineer.
- D. Remove entry ports and patch after initial cure.

GENERAL NOTES

1. Design specifications: AASHTO "LRFD Bridge Design Specifications Ninth Edition 2020".
2. Construction Specifications: State of Nevada Department of Transportation "Standard Specifications for Road and Bridge Construction 2014" except as noted in the Special Provisions for this Contract.
3. Construction type code: X028
4. "Standard Plans for Road and Bridge Construction" 2022 edition, as prepared by Nevada Department of Transportation are referenced as a part of this plan set.
5. Dimensions of the existing construction are provided in these plans to assist in the construction of the proposed bridge modifications. However, these drawings may not accurately represent field conditions. The Contractor is required to conduct necessary field investigations to verify all controlling measurements (dimensions, elevations, bearings, etc.) prior to ordering materials, preparing shop drawings or starting any construction. Submit all information to the Engineer and identify those measurements that deviate substantially from the plans and that will impact the new construction. For resolution by the Engineer. No direct payment will be made for obtaining field measurements.
6. Concrete: All new concrete shall be Class D Modified (Major) f'c=4.0 ksi @ 28 Days. Lightweight concrete shall not be used. Concrete construction joints designated as a "permissible joint" or as an "optional construction joint" may be incorporated into the construction at the contractor's option. Joints designated as a "construction joints" are considered mandatory and shall be incorporated into the construction unless otherwise approved in writing by the bridge design Engineer.
7. Reinforcing Steel: all reinforcing shall be ASTM A615 or A706 grade 60. dimensions relating to bar spacing are center to center. Bending dimensions are from out to out of the bars. Bar sizes three (3) to nine (9) are indicated by the first number of the mark; ten (10) or larger by the first two numbers. bar marks with a letter "D" suffix indicate bars that shall be doweled. Sizes four (4) and five (5), when considered as bars to control temperature, shrinkage and distribution stresses by the Engineer, may be adjusted upon concurrence and approval of the Engineer.
8. Doweled Reinforcing: reinforcing to be doweled into the existing concrete shall be epoxied in drilled holes with a maximum diameter equal to the bar diameter plus ⅛". Holes shall be cleaned with compressed air after drilling. Care shall be taken to avoid damaging existing reinforcing. minimum embedment shall be 12" unless shown otherwise. See standard plan CH-9 for additional information.
9. See roadway sheets for riprap locations.
10. See standard plans for structure excavation and backfill diagrams.

QUANTITIES

ITEM No.	ITEM DESCRIPTION	UNIT	QUANTITY
202 0125	REMOVAL OF PORTION OF BRIDGE	LS	1
203 0690	GEOTEXTILE (CLASS 1)	SQYD.	184
203 0720	GEOGRID	SQYD.	55
206 0110	STRUCTURE EXCAVATION	CUYD.	132
207 0110	GRANULAR BACKFILL	CUYD.	62
502 0390	REPAIR CONCRETE STRUCTURE	FA	1
502 0405	CRACK REPAIR	LINFT.	48
502 0970	CLASS D CONCRETE, MODIFIED (MAJOR)	CUYD.	32
505 0100	REINFORCING STEEL	POUND	6,233
505 0110	REINFORCING STEEL (DOWELED)	POUND	113
610 0470	RIP RAP BEDDING (CLASS 300)	CUYD.	74



ELEVATION

ORIGINAL CONTRACT: 1607
MODIFIED CONTRACT: 3187

STRUCTURAL DESIGN DIVISION
DESIGNED BY: G. AUFDERHEIDE / C. BELL
DRAWN BY: JESUS L. HERNANDEZ
PRINCIPAL: NATHANIEL MANGOBA

STATE OF NEVADA
DEPARTMENT OF TRANSPORTATION

US 95 - CL 50.31
TRIPLE 10' x 4' R.C.B.
ELDORADO LAKE

B-1516 N&S

DATE : 8/5/2024